

1. When we choose the first city to visit, we have four choices, since there are four cities and we haven't visited any of them yet. So the total number of ways to choose the first city is 4.
2. When we choose the second city to visit, we have three choices, since there are four cities total and we've already visited one of them. So the total number of ways to choose the first two cities is $4 \times 3 = 12$.
3. When we choose the third city to visit, we have two choices, since there are four cities total and we've already visited two of them. So the total number of ways to choose the first three cities is $4 \times 3 \times 2 = 24$.
4. When we choose the fourth city to visit, we have one choice, since there are four cities total and we've already visited three of them. So the total number of ways to choose all four cities is $4 \times 3 \times 2 \times 1 = 24$.

You should've noticed the pattern here: when we have N cities to visit, the total number of possible itineraries is $N \times (N-1) \times (N-2) \times \dots \times 3 \times 2 \times 1$, otherwise known as $N!$ (" N factorial"). The factorial function grows incredibly fast: while $3!$ is only 6 (which we can brute force without even using a computer), we find that $10!$ is over 3 million (easy enough to brute force on a modern computer), and $18!$ is over 6 quadrillion, $25!$ is over 15 septillion, and $35!$ and above starts to push the edge of what's possible to brute force on today's technology given the current expectation for the longevity of the universe.

This phenomenon is called *combinatorial explosion*. Combinatorial explosion doesn't have a rigorous mathematical definition, but it refers to cases like this, in which apparently small sets can, when considered in combinations and permutations, lead to a number of possible choices far beyond the size of the original set and beyond any size that we know how to work with using brute force.

The number of possible itineraries that connect the 90 zip codes in Rhode Island, for example, is much larger than the estimated number of atoms in the universe, even though Rhode Island is much smaller than the universe. Similarly, a chess board can host more possible chess games than the number of atoms in the universe despite the fact that a chess board is even smaller than Rhode Island. These paradoxical situations, in which the nearly infinite can spring forth from the assuredly bounded, make good algorithm